

Supplemental Material for “Wigner Resolution from Action Capacity”

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(Dated: 2026-06-02)

S1. A DE GOSSON–ZUREK–MULLOY DICTIONARY

The Letter fixes one notation. The same phase-space geometry appears in Zurek’s account of sub-Planck structure [1] and in de Gosson’s symplectic treatment of indeterminacy [2–7], each under different symbols and different normalizations. This section maps the Letter’s symbols onto theirs. The Letter sets the entries. Zurek and de Gosson reach well past the Letter, into decoherence, chaos, Donoho–Stark and Hardy inequalities, Mahler volume, and the John ellipsoid. Only the part of each work that the Letter’s symbols name is recorded here.

S1.1. Conventions and the convention bridge

The three works measure the same widths with three conventions. The Letter writes the covariance semi-axes as

$$\Delta x = \sqrt{2} \sigma_x, \quad \Delta p = \sqrt{2} \sigma_p, \quad (\text{S1})$$

so that the Heisenberg cell is the state’s covariance ellipse and its area equals the symplectic capacity. Zurek writes the widths as standard deviations, $L \simeq \sigma_x$ and $P \simeq \sigma_p$, defined from $L^2 \simeq \langle x^2 \rangle - \langle x \rangle^2$ and $P^2 \simeq \langle p^2 \rangle - \langle p \rangle^2$. The two scales differ by a fixed factor,

$$\Delta x = \sqrt{2} L, \quad \Delta p = \sqrt{2} P. \quad (\text{S2})$$

The Letter carries the area factor π inside every cell, so the quantum of action reads $h/2 = \pi \hbar$. Zurek forms bare products, $A \simeq L \times P$, without the π . de Gosson measures a cell by its symplectic capacity c , normalized so that a quantum blob has $c = \pi \hbar = h/2$. Translating the action capacity across the three,

$$A = \pi \Delta x \Delta p = c(\Omega_\Sigma) = 2\pi \sigma_x \sigma_p = 2\pi (L \times P), \quad (\text{S3})$$

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where Ω_Σ is de Gosson's covariance ellipsoid and $L \times P$ is Zurek's action.

One symbol collides across the three works and is worth flagging. In the Letter X and P are the scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$, and X_θ, P_θ are the rotated quadratures. In Zurek P is the momentum width and L the position width. In de Gosson $X \subset \mathbb{R}_x^n$ and $P \subset \mathbb{R}_p^n$ are convex bodies, the position and momentum data clouds. The dictionary below keeps each work in its own column so the three uses do not mix.

Zurek fixes these widths only to order of magnitude. He identifies the support widths L and P with the effective extent of the state, $L^2 \approx \langle x^2 \rangle - \langle x \rangle^2$, forms the bare action $A \approx L \times P$ without the area factor π , and writes the sub-Planck scale as $a \approx \hbar^2/A$. His interference relations $\delta_x = \hbar/P$, $\delta_p = \hbar/L$, and his tile area, Eqs. (7), (8), and (12), are exact once L and P are set, yet inherit the order-unity looseness of the support widths. In the Letter's covariance variables the same relations hold up to the fixed factors of Eqs. (S2) and (S3), $\sqrt{2}$ between the widths and 2π between the actions. The Letter writes $\sim$ for this reason and reserves equality for the relations symplectic polar duality fixes exactly.

S1.2. The dictionary

Tables S1 and S2 pair each symbol of the Letter with its counterpart in the two source works. The prose entries of Sec. S1 S1.3 expand the rows that carry the Letter's central relations, writing each relation out in all three notations.

S1.3. Entry by entry

a. Wigner function. The three works share one object. The Letter writes $W(x, p) = \frac{1}{\pi\hbar} \int \psi^*(x+s)\psi(x-s)e^{2ips/\hbar} ds$. Zurek's Eq. (2) is the same transform with the integration variable $y = 2s$ and the prefactor $1/2\pi\hbar$. de Gosson's Eq. (33) is the n -dimensional form $W\psi(z)$ on $\mathbb{R}^{2n}$. All three integrate to one and return the marginals $|\psi(x)|^2$ and $|\hat{\psi}(p)|^2$.

b. Capacity semi-axes and the action capacity. The Letter's Heisenberg cell is $A = \pi \Delta x \Delta p \geq h/2$. Zurek's classical action is the bare product $A \simeq L \times P$ of the support widths, Eq. (4). de Gosson's covariance ellipsoid Ω_Σ carries the same content as a symplectic capacity. His Theorem 14 states that the quantum condition $\Sigma + \frac{i\hbar}{2}J \geq 0$ holds if and only if $c(\Omega_\Sigma) \geq \pi\hbar$ for every symplectic capacity c , which is if and only if Ω_Σ contains a quantum blob. The Letter's floor $A \geq h/2$ is this capacity bound, since $h/2 = \pi\hbar$ and $c(\Omega_\Sigma) = \pi \Delta x \Delta p = A$. The conversion to Zurek's product is

This Letter (Mulloy)	Zurek [1]	de Gosson [2–4]
Position, momentum x, p	x, p	$x \in \mathbb{R}_x^n, p \in \mathbb{R}_p^n$
Wigner function $W(x, p)$	$W(x, p)$, Eq. (2)	$W\psi(z)$, Eq. (33)
Capacity semi-axis (position) L , with $L^2 \simeq \langle x^2 \rangle - \langle x \rangle^2$ $\Delta x = \sqrt{2} \sigma_x$		half-width of $X = \Pi_x \Omega_\Sigma$
Capacity semi-axis (momentum) P , with $P^2 \simeq \langle p^2 \rangle - \langle p \rangle^2$ $\Delta p = \sqrt{2} \sigma_p$		half-width of $P = \Pi_p \Omega_\Sigma$
Heisenberg cell, action capacity classical action $A \simeq L \times P$, covariance ellipsoid Ω_Σ , $c(\Omega_\Sigma) \geq \pi \hbar$, $A = \pi \Delta x \Delta p \geq \hbar/2$	Eq. (4)	Thm. 14
Resolution scale (position) $\delta x = \hbar/P \simeq \sqrt{2} \delta x$, Eq. (7) $\hbar/\Delta p$		$B_P(\Delta p)^\hbar = B_X(\hbar/\Delta p)$, Eq. (7)
Resolution scale (momentum) $\delta p = \hbar/L \simeq \sqrt{2} \delta p$, Eq. (8) $\delta p = \hbar/\Delta x$		$B_X(\Delta x)^\hbar = B_P(\hbar/\Delta x)$, Eq. (7)
Quantum blob a_θ , action $\hbar/2$	minimum-uncertainty Gaussian	Gaus- $QB(S) = S(B^{2n}(\sqrt{\hbar}))$, $c = \pi \hbar$, Eq. (26)
Principal blobs $a_{\theta=0}, a_{\pi/2}$	axis interference scales δ_x, δ_p	diagonal case $S = M_L$; John($X \times X^\hbar$), Thms. 10, 12

TABLE S1. Dictionary, part 1: core geometry. The Letter’s symbols and their counterparts in Zurek [1] and de Gosson [2–4]. Equation and theorem numbers in the Zurek column refer to [1]; those in the de Gosson column refer to [2]. Convention factors relating the columns are collected in Sec. S1 S1.1, Eqs. (S2)–(S3). Because Zurek measures with σ and the Letter with $\sqrt{2}\sigma$, his resolution scales δ_x, δ_p exceed the Letter’s $\delta x, \delta p$ by the fixed factor $\sqrt{2}$. In de Gosson’s column $B_X(R)$ and $B_P(R)$ are balls of radius R in position and momentum space and $X^\hbar$ is the $\hbar$ -polar dual of X .

Eq. (S3), $A = 2\pi (L \times P)$.

c. Resolution scales and $\hbar$ -polar duality. This is the relation the dictionary turns on. The Letter writes

$$\delta x = \frac{\hbar}{\Delta p}, \quad \delta p = \frac{\hbar}{\Delta x}. \quad (\text{S4})$$

Zurek reaches the same scales for the smallest structures of the Wigner function, $\delta_x = \hbar/P$ in his Eq. (7) and $\delta_p = \hbar/L$ in his Eq. (8). de Gosson reaches them as $\hbar$ -polar duality. For an interval of

This Letter (Mulloy)	Zurek [1]	de Gosson [2–4]
Quorum cell $\tilde{a} = \pi\delta x\delta p$	sub-Planck action $a \simeq \hbar^2/A$, $\hbar$ -polar dual $A^\hbar$; saturated pair $X \times \pi\hbar^2/(\Delta x\Delta p)$	Eq. (3); tile $(2\pi\hbar)^2/A$, $X^\hbar$ Eq. (12)
Reciprocity $\tilde{a}A = (h/2)^2$, chain $a \simeq \hbar/N$, $N = A/2\pi\hbar$ $A \supseteq a_\theta \supseteq \tilde{a}$		biduality $(X^\hbar)^\hbar = X$, antimonotone, Eqs. (4),(5)
Quantum kernel K_θ (Wigner Husimi smoothing erases function of a_θ)	(Methods)	a $W\psi_{W,Y} = (\pi\hbar)^{-n}e^{-Gz \cdot z/\hbar}$, Eq. (41)
Convolution $\widetilde{W}_\theta = W * K_\theta$	not used	smoothing toward a non-negative distribution [5, 8]
Portrait $\widetilde{W} = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta d\theta \geq 0$	not used	not used
Quadrature rotation X_θ, P_θ , angle θ	an- N, S, E, W of the compass	symplectic rotation $R \in Sp(n) \cap O(2n)$; covariance $W(\hat{S}^{-1}\psi) = W\psi \circ S$, Eq. (34)
Scaled coordinates $X = x/\Delta x$, $P = p/\Delta p$; rigid ellipse $\hat{a}_\theta$	not used	normalization to $B^{2n}(\sqrt{\hbar})$ via M_L
Saturation $A = h/2$ (vacuum), smallest scale $a \rightarrow \hbar$ $a_\theta = \tilde{a} = A$		unique blob, $c = \pi\hbar$, saturated $X^\hbar = P$, Thm. 14

TABLE S2. Dictionary, part 2: derived and dynamical quantities. Conventions and column sources are as in Table S1.

half-width R the dual is the interval of half-width $\hbar/R$, his Lemma 1, Eq. (7),

$$B_P(\Delta p)^\hbar = B_X\left(\frac{\hbar}{\Delta p}\right) = B_X(\delta x), \quad B_X(\Delta x)^\hbar = B_P\left(\frac{\hbar}{\Delta x}\right) = B_P(\delta p). \quad (\text{S5})$$

The three statements coincide once the widths are converted by Eq. (S2). Because Zurek measures with σ and the Letter measures with $\sqrt{2}\sigma$, the Letter's resolution is finer than Zurek's nominal scale by the same fixed factor, $\delta x = \delta_x/\sqrt{2}$.

d. Quantum blob. The Letter's a_θ is a de Gosson quantum blob of action $h/2$, deforming at fixed action across θ . de Gosson's definition is $QB(S) = S(B^{2n}(\sqrt{\hbar}))$ for $S \in Sp(n)$, his Eq. (26), with volume $\frac{1}{n!}(h/2)^n$ and symplectic capacity $c = \pi\hbar$. In one freedom this is an ellipse of area $\pi\hbar = h/2$, the Letter's a_θ . Zurek's building block is the round minimum-uncertainty Gaussian, the

coherent-state cell of area of order $\hbar$ from which his cat and compass states are assembled.

e. Principal blobs. The Letter's two principal members are $a_{\theta=0} = \pi \Delta x \delta p = \hbar/2$ and $a_{\pi/2} = \pi \delta x \Delta p = \hbar/2$. In de Gosson these are the diagonal, axis-aligned case. His projection theorem, Theorem 10, gives a saturated polar pair, $(\Pi_X QB(S))^\hbar = \Pi_P QB(S)$, exactly when $S = M_L$, and his Theorem 12 identifies the unique blob inscribed in $X \times X^\hbar$ as the John ellipsoid. Zurek's interference scales δ_x, δ_p are these principal members read off the coordinate axes.

f. Quorum cell, polar dual, reciprocity. The Letter's quorum cell is $\tilde{a} = \pi \delta x \delta p = \pi \hbar^2 / (\Delta x \Delta p)$, the symplectic polar dual of the Heisenberg cell, with the reciprocity $\tilde{a} A = (h/2)^2$ and the chain $A \supseteq a_\theta \supseteq \tilde{a}$. de Gosson's $\hbar$ -polar dual $A^\hbar$ is the same construction, with biduality $(X^\hbar)^\hbar = X$ and antimonotonicity, his Eqs. (4) and (5). The reciprocity follows from the interval duals of Eq. (S5), since $\tilde{a} A = \pi^2 (\delta x \Delta p) (\delta p \Delta x) = \pi^2 \hbar^2 = (\pi \hbar)^2$. Zurek's sub-Planck action is $a \simeq \hbar^2 / A$, his Eq. (3), with the compass tile $a = (2\pi \hbar)^2 / A$ of his Eq. (12). In Zurek's widths the Letter's quorum cell is $\tilde{a} = (\pi/2) \hbar^2 / (L \times P)$. He also records the count $N = A/2\pi \hbar$ of orthogonal states and the scale $a \simeq \hbar/N$.

g. Quantum kernel, convolution, portrait. The Letter's quantum kernel K_θ is the Wigner function of the blob a_θ , a centered Gaussian of action $\hbar/2$. de Gosson's Eq. (41) gives the Wigner function of a generalized Gaussian, $W\psi_{W,Y}(z) = (\pi \hbar)^{-n} e^{-Gz \cdot z / \hbar}$, which is the kernel written through its covariance. Convolution with a Gaussian of action $\hbar/2$ returns a non-negative distribution, the route through Cartwright [8] and de Gosson's Husimi–Wigner cells [5]. The Letter's per-angle convolution is $\widetilde{W}_\theta = W * K_\theta$ and the portrait is the family integral $\widetilde{W} = \frac{1}{\pi} \int_0^\pi \widetilde{W}_\theta d\theta \geq 0$. Zurek notes the complementary fact that the Husimi function, the Wigner function smoothed on a full Planck scale, erases the sub-Planck scale a . The portrait sits between the two, resolved at $\tilde{a}$ rather than at $\hbar$.

h. Quadrature rotation. The Letter's rotated quadratures are $X_\theta = X \cos \theta + P \sin \theta$ and $P_\theta = -X \sin \theta + P \cos \theta$, the homodyne quadratures at phase θ that orient the blob and the kernel. de Gosson carries the same rotation as a symplectic rotation $R \in Sp(n) \cap O(2n)$ and as the metaplectic covariance of the Wigner function, $W(\hat{S}^{-1}\psi) = W\psi \circ S$, his Eq. (34). The family integral over θ is the tomographic sweep [9, 10]. Zurek's compass orients its lobes north, south, east, and west, fixing the two axis directions of the interference grid.

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